

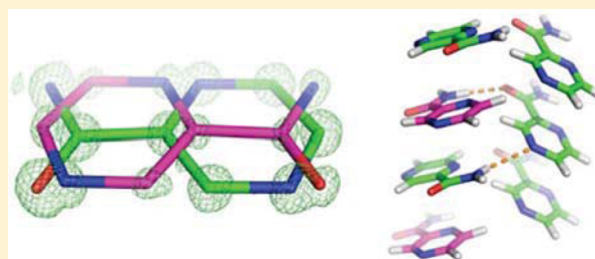
Polymorph Stability Prediction: On the Importance of Accurate Structures: A Case Study of Pyrazinamide

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ABSTRACT: We reproduce the experimentally observed polymorph stability of the tetramorph pyrazinamide using periodic density functional theory (DFT) calculations augmented with an empirical dispersion term. This was only possible using highly accurate experimental structures, including a model of a disordered polymorph. Further, we demonstrate that a large basis set and very tight integration criteria are needed in the calculations. This study highlights some of the present challenges in crystal structure prediction methods: to obtain sufficiently accurate trial structures and to take into account the possibility of crystal disorder.



INTRODUCTION

Why do polymorphs present a challenge to crystal structure predictors,¹ in spite of the recent success in predicting systems with only one known structure?² The advancement of crystal structure prediction (CSP) relies on the ability to assess the crystal energies (enthalpies). However, in the case of polymorphic systems, these energy differences may be too small to be the determining factor for the outcome of crystallization. Other factors, such as vibrational entropies, may alter the thermodynamic stability, i.e., the free energy, of the crystals,^{3–5} and kinetics of nucleation and crystal growth governed by the crystallization conditions, may become important.⁶ These factors are not considered in present-day structure prediction procedures.

The accuracy of the present schemes to calculate the crystal enthalpies can also be part of the problem. This may be evaluated using results from periodic density functional theory (DFT) calculations based on highly accurate X-ray diffraction structures compared with experimental data. In the present work, the accuracy is solely determined by the ability to reproduce the relative stability of the polymorphic forms of the test molecule.

We selected the tetramorphic system pyrazinamide as a test case, because pyrazinamide (Figures 1 and 2) has been investigated intensely as an antituberculosis drug. Recently, two groups have independently studied the stabilities by calorimetric measurements, X-ray diffraction, spectroscopy, and visual observation of phase transformations. The polymorphs are enantiotropically related and show a large range of solid-state transformations (Figure 3) upon heating or mechanical stress. These transformations have given a good understanding of the relative stabilities of the polymorphs at all temperatures.^{7,8}

Besides the known physical properties of the polymorphs, pyrazinamide is computationally fairly easy to handle because

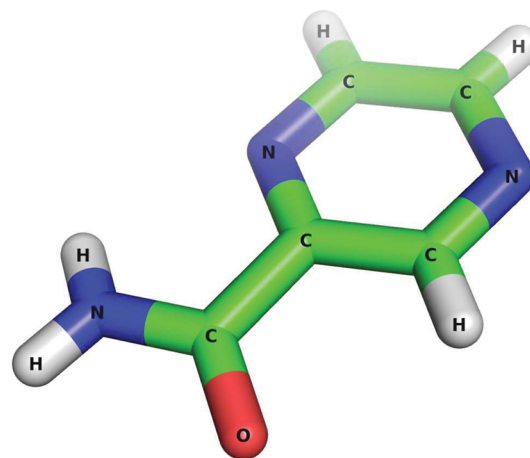


Figure 1. The molecular structure of pyrazinamide.

the molecule is small and rigid and does not contain heavy elements. This allows us to test different computational routines, which are used to determine the order of stability of the four polymorphs and to compare the results with experimental data.

The structures of all four polymorphic forms have previously been determined by X-ray diffraction, albeit with varying precision, and at different temperatures. As part of this investigation, very precise diffraction experiments were performed at the same temperature for all forms.

Crystallographic Details. The four polymorphs of pyrazinamide were obtained by crystallization from various

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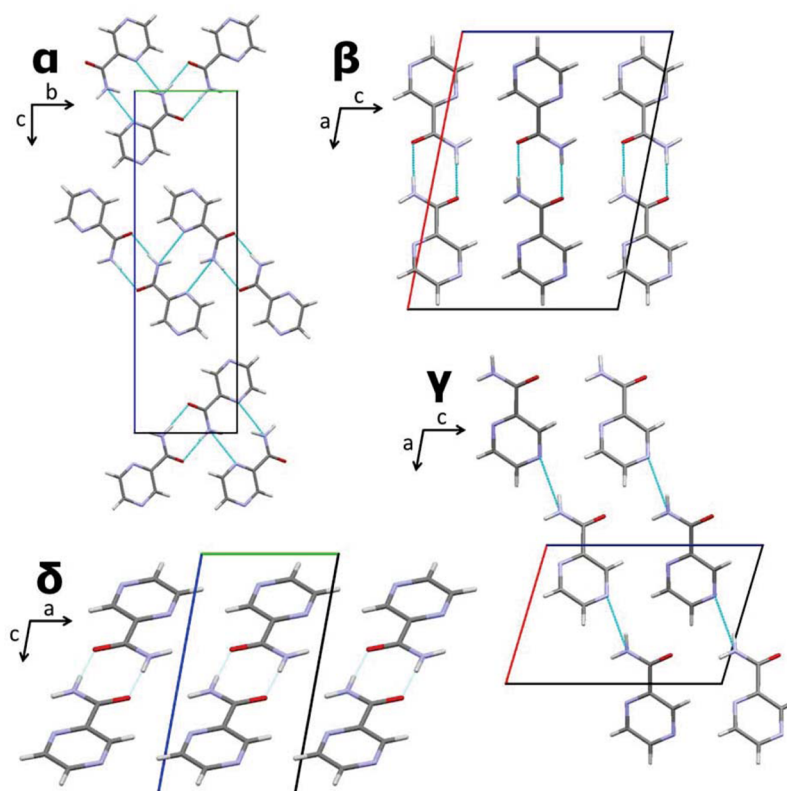


Figure 2. Pyrazinamide packing diagrams. Alpha, beta, and delta polymorphs form dimers, while the gamma form orders in linear chains.

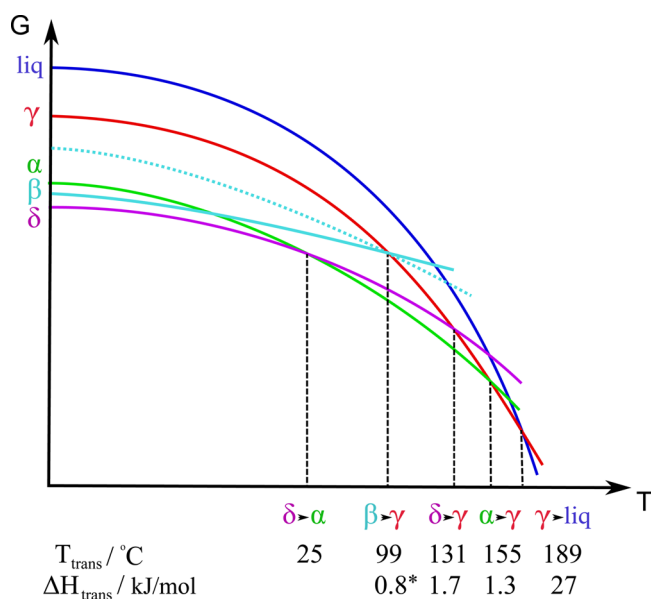


Figure 3. Schematic drawing of the free energies of the polymorphs and their temperature dependence. Thermodynamic data based on DSC measurements.⁷ *Data for the beta form are approximate since the beta form could only be crystallized concomitantly with the gamma form. No uncertainties on the thermodynamic data were given by Cherukuvada and co-workers. The free energy curve for the beta form as proposed by Cherukuvada is shown as a dotted cyan line.

solvents, in accordance with the procedures outlined by Castro et al.⁸

The alpha, beta, and delta polymorphs crystallized from the saturated solution by evaporation of the solvent at room temperature, the respective solvents being water, chloroform,

and dichloromethane. Crystals of the gamma form produced from sublimation crystallization were kindly provided by Castro and co-workers. However, the diffraction quality of the crystals obtained by sublimation was not sufficient to obtain a highly accurate result. Instead, crystals of the alpha form were slowly heated in an oven to 160 °C and in what appears to be a single crystal to single crystal transformation turned to the gamma form. These crystals were of excellent diffraction quality.

The kinetic stability of the four polymorphs have been established by Castro et al.⁸ and Cherukuvada et al.⁷ based on differential scanning calorimetry (DSC) measurements, visual microscopy, and Raman spectroscopic measurements. At lower temperatures, all four polymorphs are kinetically stable, while at elevated temperatures the alpha, beta, and delta transform to the gamma phase. It was therefore possible to redetermine the structures of the four polymorphs at 122 K using a Nonius KappaCCD diffractometer equipped with Mo radiation and a graphite monochromator. The crystallographic data are summarized in Table 1.

Packing diagrams of the resulting structures are given in Figure 2. The packing for the alpha, beta, and delta forms consists of dimers of molecules, where a double hydrogen bond connects the amide groups. In contrast, the gamma form is ordered in linear chains of molecules, with the amide groups donating to a nitrogen of the pyrazine ring. There is a considerable amount of disorder in the gamma form, vide infra.

The structures confirm previous work, however with a much better resolution and precision of the data: The temperature was calibrated to exactly 122 K by observing the phase transition of potassium dihydrogen phosphate. Data were collected to a resolution better than 0.5 Å, with a redundancy better than 3. As a further test, the structures were subjected to Hirshfeld's rigid bond test:⁹ For bound atoms of nearly equal

Table 1. Crystallographic Data^a

	α	β	γ	δ
space group	$P2_1/n$	$P2_1/c$	Pc	$P\bar{1}$
Z	4	4	2	2
Z'	1	1	1	1
cell length/Å	$a = 3.6690(4)$ $b = 6.7110(8)$ $c = 22.518(3)$	$a = 14.3280(3)$ $b = 3.6330(13)$ $c = 10.6280(17)$	$a = 7.1730(17)$ $b = 3.6500(13)$ $c = 10.686(2)$	$a = 5.1280(14)$ $b = 5.7080(12)$ $c = 9.857(4)$
cell angles/°	$\beta = 92.0470(2)$	$\beta = 101.169(7)$	$\beta = 106.487(13)$	$\alpha = 97.45(3)$ $\beta = 98.03(3)$ $\gamma = 106.55(2)$
temperature/K	122.4(5)	122.4(5)	122.4(5)	122.4(5)
measured reflections	49596	47668	17605	15213
unique reflections	5797	5683	5658	4988
R_{int}	0.0580	0.0394	0.0420	0.0742
R	0.0740	0.0644	0.0875	0.0525
wR	0.1337	0.1217	0.1219	0.1269
goodness of fit	1.076	1.049	1.022	1.090
Δ_{bond}	0.0003	0.0004	0.0004	0.0003

^aAll data were collected on a Nonius KappaCCD at 122 K using Mo radiation. Δ_{bond} is the root mean square difference in mean square displacements (Hirshfelds rigid bond test⁹): $\Delta_{\text{bond}} = (\sum \delta_{ij}^2/N)^{1/2}$ where δ_{ij} is the difference in mean square displacements in the bond direction of the bonded atoms i and j .

mass (like C, N, and O), the mean square displacement in the bond direction should be similar, with a difference of less than 10^{-3} Å^2 . All the structures pass the test, in contrast to the previously reported structures. The average differences for all bonds in the structures are reported in Table 1.

Computational Procedure. It is well-known that crystal structures from X-ray diffraction experiments are of good accuracy, except with respect to the positions of hydrogen atoms. We used our accurate structures as a starting point for the geometry optimizations and energy assessments. Three different optimization approaches were tested to calculate the relative crystal energies.

(I) A full geometry optimization of all cell parameters and atomic coordinates. This approach mimics the procedure typically used in crystal structure prediction routines.

(II) A geometry optimization of the atomic coordinates, while the cell parameters were adopted from the 122 K measurement.

(III) A geometry optimization of the hydrogen positions, while the atomic positions of the heavier elements and cell parameters are constant (122 K). Only the position of the hydrogen atoms were optimized to circumvent the well-known problem of determining H nuclear positions in X-ray crystallography.

The energies were calculated using the CRYSTAL09 code.¹⁰ Calculations were performed at the HF level as well as using the hybrid DFT method B3LYP.¹¹ We employed Poples 6-31G(d,p) basis set, as well as a slightly modified version of Ahlrichs VTZ basis set with polarization functions.¹² The modified Ahlrichs VTZ basis, denoted VTZ-mod, is the original basis with a rescaling of the largest diffuse s-function on the hydrogen atoms (from 1.00 to 1.10). This slight contraction was done to prevent the system from reaching an artificial conductive state, which is due to numerical errors in the calculations.¹³ To improve the long-range interaction of the B3LYP functional, Grimmes dispersion correction, denoted as B3LYP-D, was employed.^{14,15}

The truncation parameters, TOLINTEG, were set to 6 6 6 6 16, the first two being the overlap and penetration threshold for

the Coulomb integrals, the third being the overlap threshold for the HF exchange integrable, and the last two being pseudo-overlaps in the HF exchange-series. Eigenvalue level shifting (LEVSHIFT) was used at all times; the smaller basis set 6-31G(d,p) was shifted 0.3 hartree, while the larger, VTZ-mod, was shifted 0.6 hartree. The numerical integration of the DFT exchange-correlation contribution involved a grid pruning scheme.¹⁶ A (75 974) grid was used that contains 75 radial points and a variable number of angular points, with a maximum of 974 on the Lebedev surface in the most accurate integration region (XLGRID option in Crystal09). The standard integration grid was not sufficient; differences of up to 1.25 kJ/mol/molecule were observed between the standard grid and the employed XLGRID. The shrinking factors in the reciprocal space, SHRINK, were set to 4 and 4. The mixing of the Fock and Kohn–Sham matrix was set to 30%.

The basis set superposition error (BSSE) was calculated using the counterpoise correction¹⁷ as implemented in the Crystal09 program:

$$E_{\text{BSSE}} = E_{\text{mol+ghost}} - E_{\text{molecule}} \quad (1)$$

where $E_{\text{mol+ghost}}$ is the energy of the molecule including ghost orbitals to a distance of 4 Å, and E_{molecule} is the energy of a single molecule. The model of the disordered gamma form represents a special case with four molecules in the asymmetric unit. Instead of on a single molecule, the energy of all four molecules at infinite distance (the MOLSPLIT routine in Crystal09) is calculated and compared to the energies of the four molecules surrounded by ghost orbitals to 4 Å. Thus, the total BSSE of the asymmetric unit is obtained. The total BSSE is divided by 4 to obtain an average BSSE per molecule which is comparable to the molecular BSSE of the other forms:

$$E_{\text{BSSE}} = \frac{\sum_1^4 E_{i' \text{ th mol+ghost}} - E_{\text{mol split}}}{4} \quad (2)$$

The BSSE correction was applied to the optimized structures and was not part of the optimization routine.

The Disordered Gamma Form. The gamma form presents a special problem. A structural refinement using the standard independent atom model results in high refinement residuals, pointing toward ill-behaved structural features. Further, the quality of the intramolecular geometry and the anisotropic displacement parameters in the structure of Castro and co-workers indicated that the quality and the model could be improved. A careful inspection of the residual electron density map revealed a second position of the gamma molecule (Figure 4). Similar observations were done by Cherukuvada et

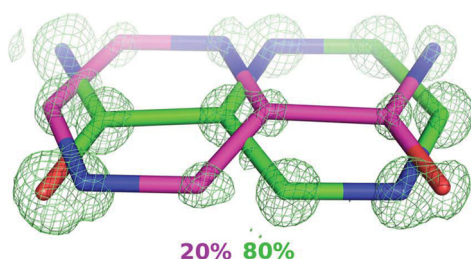


Figure 4. Electron density map of the gamma form, showing the model of C, N, and O atoms of the two orientations constituting the disorder in the crystal. $|2Fo - Fc|$ map with contours at the 1.0σ level.

al.⁷ Including this second molecule in the model as disorder (SHELXL¹⁸ refinement using PART instructions) shows that the site has an occupation of about 20%. This occupation was apparently independent of the crystallization conditions. The residual density can be the consequence of disorder, or it can be due to domain-intergrowth of two distinctly different polymorphic forms, having similar unit cell parameters (Figure 5). Two different approaches were used to test these hypotheses: one crystallographic, the other computational.

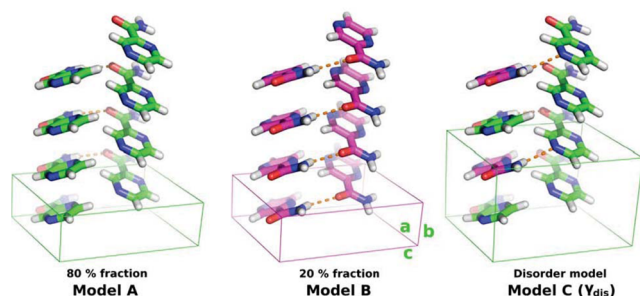


Figure 5. Three hypothetical models of the gamma crystal. Left: The main (80%) form, without disorder. Middle: A model composed solely of the minor (20%) rotated form. Right: A model of the disorder, incorporating the rotated molecule, as indicated by the peaks in the residual density. This model was used to compute the energy of the rotated molecule in the matrix of molecules of the main form.

First, the Jana program¹⁹ was used to test whether it was possible to distinguish between a randomly disordered crystal or a situation of two merohedrally intergrown polymorphic domains, similar to a test for twinning. Whereas merohedral twinning results in diffraction intensities that are the sum of the intensities of the individual domains, disorder gives rise to intensities that are proportional to the square of the added structure factors of the (weighted) disordered parts. The two different models refined to $wR(F^2)$ values of 5% and 17% respectively, strongly supporting the hypothesis that the gamma crystal is randomly disordered. This conclusion was further

Table 2. Differences in Cell Dimensions and Energy-Difference (kJ/mol/molecule) of the Alpha, Beta, and Delta Form from Different Crystallographic Experiments^a

	α	β	δ
a	3.6147(4)	14.372(7)	5.1186(10)
	3.6690(4)	14.3280(3)	5.1280(14)
b	6.7384(8)	3.711(3)	5.7053(11)
	6.7110(8)	3.6330(13)	5.7080(12)
c	22.464(3)	10.726(5)	9.857(2)
	22.518(3)	10.6280(17)	9.857(4)
α angle	90	90	97.46(3)
	90	90	97.45(3)
β angle	92.499(2)	101.92(5)	98.17(3)
	92.0470(2)	101.169(7)	98.03(3)
γ angle	90	90	106.47(3)
	90	90	106.55(2)
B3LYP/6-31G(d,p)			0.63
B3LYP-D/6-31G(d,p)			0.72
B3LYP/VTZ-mod	2.02	−6.1	0.91
B3LYP-D/VTZ-mod	−0.25	3.40	0.17

^aThe unit cell dimensions of the literature study are given as the first row (100 K, Cherukuvada et al.⁷), whereas the dimensions found in the present study (122 K) are given as the second row. The difference at various levels of theory was obtained using the experimental structures from the literature and compared to the experimental structures (122.4 K) determined in the present study. H positions were optimized in accordance with procedure III.

supported by inspection of diffraction images that show no signs of diffuse scattering.

Second, for the computational test three different crystallographic models were constructed. Figure 5: model A is a crystal without disorder, only composed of the 80% component of the disorder-refined model. Model B is composed solely of the 20% component. Model C was made to make an estimated model of the disordered crystal. The symmetry was reduced from $P21/c$ to $P1$, and the unit cell doubled along the b -axis. One out of the four molecules were in the geometry corresponding to model B, and the three other according to model A. The Crystal09 program was used to compute the crystal energies of the three models using approach III (see the computational section), using B3LYP/VTZ-mod. The energy of model B was 12 kJ/mol/molecule higher than model A. Model C was only 3 kJ/mol higher in energy than model A. The relatively high energy of model B suggests that this form is not likely to be observed. A crystal composed of intergrown domains of A and B is therefore not likely, and the lower energy of model C favors the conclusion that the crystal is randomly disordered.

To conclude, both the crystallographic and computational models support the hypothesis that the gamma form is randomly disordered in the ratio of 1:4 between the two orientations. The constructed model C is an approximation. A more thorough treatment of disorder would involve studying the symmetry adapted ensemble of structures;²⁰ for the present purpose we feel confident that we have created a good approximation of the disordered structure, and we used model C to compute the energies of the gamma form in the results presented below. In the tables, this model is named γ_{dis} to signify that we have taken the disorder into account in the calculations. The residual densities of the three other polymorphic forms were inspected, but in these systems there were no indications of disorder.

DISCUSSION OF COMPUTATIONAL ASPECTS

In the following section, we touch upon the different important aspects of the computations. The basis set superposition error (BSSE), dispersion correction of the DFT energies, and the accuracy of the experimental structures must all be considered in order to obtain trustworthy results.

BSSE. Basis set superposition errors were estimated using the counterpoise procedure and are listed in Table 8. Large BSSEs are observed when the smaller basis set is used (on the order of 39 kJ/mol). Besides the high absolute size of the error, the difference in BSSE between the polymorphs is about 2 kJ/mol and thus of the same order of magnitude as the expected differences in energies. This situation is greatly improved when employing the larger VTZ-mod basis where the BSSE drops to about 7.5 kJ/mol. This is still considerable, but the largest difference between polymorphs is now as low as 0.4 kJ/mol. The size of the BSSE is a reminder that we should take the computed energies *cum grano salis* and at most expect to be able to rank the energy of polymorphs in the right order. BSSE corrections were applied to the final, optimized structures, and were not part of the optimization strategy.

Dispersion Correction. The dispersion interactions are not well described by the density functional theory. A lot of work has been done to add dispersion to DFT for molecular crystals²¹ with successful applications in plane wave methods.²² Grimme¹⁴ has proposed to use an empirical dispersion correction, and his approach has been implemented in the Crystal09 program.¹⁵ The advantages and disadvantages of the DFT approach as opposed to force-field based approaches have been recently reviewed.²³

The polymorph energies of pyrazinamide with and without the Grimme correction are given in Tables 3–5. The advantage

Table 3. Approach I: Energy Determined Using B3LYP-D and B3LYP Combined with 6-31G(d,p) and VTZ-mod Basis Sets Relative to δ in kJ/mol/molecule^a

	α	β	γ_{dis}	δ
B3LYP/6-31G(d,p)	0.45	0.02	1.4	0.0
B3LYP-D/6-31G(d,p)	−0.61	−1.47	3.26	0.0
B3LYP/VTZ-mod	−2.63	−4.25		0.0
B3LYP-D/VTZ-mod	−0.89	−1.29		0.0

^aThe energies were corrected for basis set superposition error using the counterpoise procedure. The disordered gamma form γ_{dis} corresponding to model C in Figure 5, could not be optimized using the large basis set.

Table 4. Approach II: Energy Determined Using B3LYP-D and B3LYP Combined with 6-31G(d,p) and VTZ-mod Basis Sets Relative to δ in kJ/mol/molecule^a

	α	β	γ_{dis}	δ
B3LYP/6-31G(d,p)	−1.91	−4.06	−0.51	0.00
B3LYP-D/6-31G(d,p)	−0.50	−0.86	6.37	0.00
B3LYP/VTZ-mod	−1.67	−3.80	0.08	0.00
B3LYP-D/VTZ-mod	2.23	1.54	0.34	0.00

^aThe energies were corrected for basis set superposition error using the counterpoise procedure. The disordered gamma form γ_{dis} corresponds to model C in Figure 5.

of using Grimmes correction is evident for both basis sets: The delta crystal form becomes the most stable form, in accordance with experiments. The cell parameters from the fully geometry

Table 5. Approach III: Energy Determined using B3LYP-D and B3LYP Combined with 6-31G(d,p) and VTZ-mod Basis Sets Relative to δ in kJ/mol/molecule^a

	α	β	γ_{dis}	δ
B3LYP/6-31G(d,p)	−2.59	−4.51	1.78	0.00
B3LYP-D/6-31G(d,p)	−0.61	−1.47	3.26	0.00
B3LYP/VTZ-mod	−1.60	−3.42	1.30	0.00
B3LYP-D/VTZ-mod	2.65	0.76	3.57	0.00

^aThe energies were corrected for basis set superposition error using the counterpoise procedure. The disordered gamma form γ_{dis} corresponds to model C in Figure 5.

optimized structures (method I, B3LYP-D/VTZ-mod) are compared to the experimental cell parameters in Table 6. The optimized cell parameters are generally smaller than the experimental, by about 1–4%. Some of these differences can be ascribed to temperature effects, and we should expect better agreement if the experimental cell parameters were obtained at very low temperatures. The largest deviations from the experimental cell parameters can be observed for the cell vectors along the stacking direction of the pyrazine rings. This is the case for all four polymorphs. The ring stacking is characterized by dispersion forces, and the results indicate that there is still room for improvement of the dispersion correction of the DFT methods.

When the optimization is performed without dispersion correction, much larger deviations are observed. The cell volumes (Table 7) are more than 30% larger than the experimental.

Experimental Structural Accuracy. It is important to question the accuracy of the structures obtained from crystallography, as well as the structures obtained from energy optimization. We address the question by comparing crystallographic figures of merit for the different structures, as well as the energy of different experimental structures of the polymorphs. The energies were computed using procedure III, where only H atom positions are optimized, and at the B3LYP-D/VTZ-mod level of theory.

In the following discussion, please refer to Table 2. A small energy difference between the structures in the literature and the structures determined in our work is expected due to the different temperatures (100 K vs 122 K) giving slightly different unit cell dimensions. Furthermore, there is a difference in structural accuracy and precision. The data collected by Cherukuvada et al.⁷ extend to 26 degrees, perfectly adequate to obtain the crystal structure. However, given the structural accuracy needed in the energy calculations, we have more trust in the data collected for the present study. The data range extends to 52 deg — about 5 times the number of observed Bragg reflections as in the study of Cherukuvada. The larger 2θ range results in more information about the core electrons and thus the atomic positions. The increased number of reflections helps for a better data to parameter ratio and consequently a more trustworthy structure. The higher precision and accuracy manifest in lower standard uncertainties on the atomic coordinates as well as more physically acceptable anisotropic displacement parameters. These structural differences give rise to differences in the crystal energies of 0.25 kJ/mol (α), 4.0 (β), and 0.16 (δ) kJ/mol. In all cases, the interaction energies of the more accurate structures were larger, in spite of the higher temperature. This stresses the fact that very accurate

Table 6. Optimized Cell Parameters (Method I) against Experimental Results (122 K)^a

	<i>a</i> /Å	<i>b</i> /Å	<i>c</i> /Å	α /°	β /°	γ /°
alpha						
experimental	3.615	6.738	22.464	90.00	92.50	90.00
optimized	3.487	6.685	22.037	90.00	96.90	90.00
% deviation	−3.54	−0.79	−1.90		4.76	
beta						
experimental	14.328	3.633	10.628	90.00	101.17	90.00
optimized	14.111	3.480	10.314	90.00	98.34	90.00
% deviation	−1.51	−4.20	−2.96		−2.80	
gamma						
experimental	7.173	3.650	10.686	90.00	106.49	90.00
optimized	7.156	3.491	10.423	90.00	106.35	90.00
% deviation	−0.24	−4.36	−2.46		−0.13	
delta						
experimental	5.119	5.705	9.857	97.46	98.17	106.47
optimized	5.038	5.628	9.657	98.56	99.97	105.64
% deviation	−1.58	−1.35	−2.03	1.13	1.83	−0.78

^aResults are given for the B3LYP-D/VTZ-mod method. Notice that the model of the disordered gamma form (Model C in Figure 5) could not be optimized using the largest basis set, and the results are given for optimization the model consisting of the main fragment (Model A in Figure 5).

Table 7. Cell Volume in Å^{3a}

	α	β	γ_{dis}	δ
crystallographic	546.64	546.31	536.54	268.83
B3LYP/6-31G(d,p)	627.68	717.92	620.52	313.36
B3LYP-D/6-31G(d,p)	498.14	487.59	494.39	247.19
B3LYP/VTZ-mod	695.17	754.99		373.74
B3LYP-D/VTZ-mod	509.90	501.20		254.13

^aThe disordered gamma form γ_{dis} corresponds to model C in Figure 5.

Table 8. Basis Set Superposition Error (BSSE) Calculated Using the Counterpoise Scheme and the Geometries from Procedure III^a

	α	β	γ_{dis}	δ
B3LYP/6-31G(d,p)	−38.70	−38.44	−38.03	−40.86
B3LYP-D/6-31G(d,p)	−38.49	−38.43	−38.83	−40.86
B3LYP/VTZ-mod	−7.92	−7.57	−7.91	−7.60
B3LYP-D/VTZ-mod	−7.98	−7.57	−7.91	−7.50

^aEnergies in kJ/mol/molecule.

structures measured at the same temperature are crucial to obtain trustworthy results.

RESULTS AND DISCUSSION

We begin by restating the hypothesis of the present investigation: The accuracy of the schemes to calculate the crystal enthalpies used in structure prediction is not sufficient. The schemes are typically composed of three steps: (1) A large number of trial structures are generated. (2) The most likely structures are chosen. Likelihood can be based on chemical intuition, force-field calculations, densities, etc. (3) The most likely structures are further optimized, and a better guess of the crystal energy is calculated based on more force fields, or on ab initio calculations.

The experimentally observed polymorphs are most often still part of the prediction set after step 2. Step 3 is the crucial step in discriminating between observed and virtual crystal structures based on the relative energies. Ideally, the relative

energies of the observed polymorphs should match what is found experimentally.

The crystal energies derived from ab initio calculations correspond to the free energies at 0 K, if corrected for zero-point vibrational energies.²² It is not trivial to determine the relative thermodynamic stability of polymorphs at 0 K experimentally. Calorimetric measurements such as DSC are typically used to assess the stability at ambient conditions; differences in melting point and heats of fusion or sublimation can be used to assess the relative stabilities on a qualitative level. In the case of pyrazinamide, a tentative free-energy temperature diagram has been constructed based on such measurements and other observations of phase transformations (Figure 3). The relative stability based on observations of phase transitions and their enthalpies is $\delta > \alpha > \gamma$, with δ being the most stable form, and further $\beta > \gamma$. The β form is difficult to place within the order of stability at 0 K. It is difficult to isolate and has only been obtained in mixtures with the γ form. DSC experiments on a mixture of the γ and β forms⁷ suggest that the order of stability is $\delta > \beta > \alpha > \gamma$ at 0 K.

As outlined in the computational section, we have investigated a number of possible procedures to assess the relative stabilities of the four polymorphs. We have discussed issues regarding the size of basis set in relation to the BSSE, and the use of Grimmes dispersion correction. In the following discussion, we will only consider the results related to our most advanced method, using B3LYP-D/VTZ-mod. The last rows of Tables 3–5 summarize these results.

The first procedure I presented is the full optimization of the coordinates and unit cell parameters. This is very similar to the approach used in crystal structure prediction routines, which of course do not have any experimental information. This procedure does not reproduce the experimentally observed order of stability. The δ form, which is known to be the most stable form, becomes less stable than the α and β forms. Notice that the optimized unit cells are contracted as compared to the cells observed at 122 K, as we should expect due to thermal effects. The deviation from the observed cells are less than 5% for all unit cell parameters, and in many cases

much smaller. The magnitude of this deviation is in line with the typical results of crystal structure prediction routines.

Procedure II is an optimization of the atomic coordinates, while keeping the unit cell parameters at the experimental results. The resulting relative stabilities still deviate significantly from what is experimentally observed but may be improved by using the dispersion correction. When the functional is dispersion corrected the alpha and beta form are less stable than the delta form in accordance with experimental observations. The disordered gamma form has a slightly lower energy than the delta form, the opposite of what is reported from experiments.

In procedure III, we calculated the energy of the experimental structures but with an optimization of only hydrogen positions to account for the known shortening of the X–H bondlengths in X-ray crystallography. In this approach, we observe the correct order of stability. The gamma phase is by far the least stable ($E_{\gamma_{\text{dis}}} - E_{\delta} = 2.94$ kJ/mol/molecule), followed by the alpha form ($E_{\alpha} - E_{\delta} = 1.06$ kJ/mol/molecule) and beta ($E_{\beta} - E_{\delta} = 0.14$ kJ/mol/molecule). The energies were corrected for basis set superposition error but the order of stability is the same without this correction. Notice that the beta form is calculated as more stable than the alpha form at 0 K. The E – T diagram proposed by Cherukuvada et al. suggests the alpha form to be more stable than beta at 0 K; however, we do not find experimental evidence that supports this assumption: The thermodynamic data for the beta polymorph were performed on a mixture of gamma and beta form; therefore the reported transformation enthalpy⁷ from beta to gamma must be considered a lower bound. A revised E – T diagram, which we believe is in agreement with both experiments and calculations, is given in Figure 3. In comparison to the E – T diagram given by Cherukuvada and co-workers, we have set the stability of the beta form to be higher than the stability of the alpha form. The original curve proposed by Cherukuvada and co-workers is indicated as a stipled yellow line in Figure 3. Because we know that the beta form has got the lowest transformation temperature (to the gamma form) the free energy curve for beta is less steep than the curve for the other forms. This seems to indicate that the beta form has got a lower solid state entropy than the other forms.

CONCLUSION

A range of different computational approaches were tested with the aim to reproduce the order of stability of pyrazinamide polymorphism. An optimization of the entire structure, including the unit cell parameters, does not reproduce the order of stability found experimentally. However, this could successfully be achieved using the experimental structures with optimized hydrogen positions. This indicates that the precision of the optimization procedure, even when using an extensive basis set, and a DFT method including empirical dispersion correction, is not sufficiently accurate to be useful for studying the relative stability of polymorphic systems. Since the deviations observed between the optimized structure and the experimental one is similar to the deviations found in crystal structure prediction (CSP), this failure may also provide an explanation for the common failure of CSP routines to distinguish between real and virtual polymorphs.

On the other hand, the experimental structures are sufficiently accurate and may therefore be used as benchmarks

for improving the ab initio structure optimization techniques. If the accuracy of the structure optimization is to be improved, this study demonstrates that it is within reach to assess the relative stability of polymorphs at 0 K. Given the differences in cell volumes between the studied polymorphs, thermal motion can be different between these systems and should give rise to differences in vibrational entropy. Work is in progress to provide a procedure for assessing the stability at ambient temperatures, including the vibrational enthalpy and entropy in the estimation of the free energy.

In relation to structure prediction, the disordered gamma form of pyrazinamide represents a frontier research problem; it is far from trivial to predict a disordered structure.^{24,25} In the present case, we were able to mimic the experimentally observed disorder by reducing the symmetry and performing a supercell calculation with one molecule flipped. However, there is a plethora of possible disordered forms which reduces the space group symmetry. Thus, the prediction of disorder resembles the situation for predicting structures with more than one molecule in the asymmetric unit; the computational cost is too large for quantum-chemical calculations but may be treated by force-field based calculations.

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Notes

The authors declare no competing financial interest.

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